

Artificial Intelligence (CSA17) — Assessment Tool 3: Resolution Algorithm — Model Answers

Assessment Tool 4 — Resolution Algorithm Implementation (CO3)

Question 1 — Rain and Wet Ground

Let **R** = “It is raining”, **W** = “The ground is wet”.

- Premise 1: $R \rightarrow W$
- Premise 2: R (fact)
- Goal: W

CNF: $\neg R \vee W$ (from Premise 1), R (Premise 2, unit clause) **Negate the goal:** $\neg W$

Resolution steps: 1. Resolve $(\neg R \vee W)$ with (R) on $R \rightarrow W$ 2. Resolve W with the negated goal $\neg W \rightarrow$ **Empty clause** $\square$

Conclusion: The empty clause is derived, so “the ground is wet” is proved true.

Question 2 — Student Assignment Submission

Let **S** = “Rahul submits the assignment”, **M** = “Rahul receives internal marks”.

- Premise 1: $S \rightarrow M$
- Premise 2: S (Rahul submitted the assignment)
- Goal: M

CNF: $\neg S \vee M$, S **Negate the goal:** $\neg M$

Resolution steps: 1. Resolve $(\neg S \vee M)$ with (S) on $S \rightarrow M$ 2. Resolve M with $\neg M \rightarrow$ **Empty clause** $\square$

Conclusion: Goal proved — **Rahul receives internal marks.**

Question 3 — Library Membership

Let **L** = “Priya is a library member”, **B** = “Priya can borrow books”.

- Premise 1: $L \rightarrow B$
- Premise 2: L (Priya is a library member)
- Goal: B

CNF: $\neg L \vee B$, L **Negate the goal:** $\neg B$

Resolution: 1. Resolve $(\neg L \vee B)$ with (L) on $L \rightarrow B$ 2. Resolve B with $\neg B \rightarrow$ **Empty clause** $\square$ — no further clauses can be generated after this.

Final conclusion: Priya can borrow books — proved true.

Question 4 — Placement Eligibility

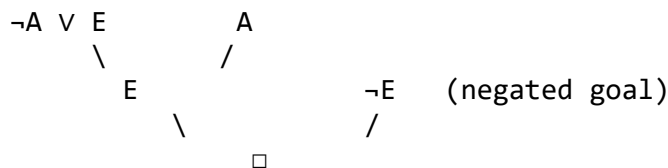
Let A = “Arun clears the aptitude test”, E = “Arun is eligible for placement”.

- Premise 1: $A \rightarrow E$
- Premise 2: A (Arun cleared the aptitude test)
- Goal: E

CNF clauses: $\neg A \vee E$, A **Negate the goal:** $\neg E$

Resolution steps: 1. Resolve $(\neg A \vee E)$ with (A) on $A \rightarrow E$ 2. Resolve E with $\neg E \rightarrow$ **Empty clause** $\square$

Resolution tree:



Conclusion: Arun is eligible for placement — proved.

Question 5 — Access Control System

Let P = “the user enters the correct password”, $Auth$ = “the user is authenticated”, Acc = “the user is granted access”.

- Premise 1: $P \rightarrow Auth$
- Premise 2: $Auth \rightarrow Acc$
- Premise 3: P (fact — correct password entered)
- Goal: Acc

CNF: $\neg P \vee Auth$, $\neg Auth \vee Acc$, P **Negate the goal:** $\neg Acc$

Resolution steps (with explanation): 1. Resolve $(\neg P \vee Auth)$ with (P) on $P \rightarrow Auth$ — the user is authenticated because the password was correct. 2. Resolve $Auth$ with $(\neg Auth \vee Acc)$ on $Auth \rightarrow Acc$ — since the user is authenticated, access is granted. 3. Resolve Acc with the negated goal $\neg Acc \rightarrow$ **Empty clause** $\square$ — the contradiction confirms the negation is impossible, so the goal must hold.

Final conclusion: The user is granted access — proved true via the Resolution Algorithm.